



SAVEETHA
UNIVERSITY

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SAVEETHA
SCHOOL OF ENGINEERING

Engineer to Excel

► **TOPIC**

SMART RAILWAY RESERVATION WITH EFFICIENT ENERGY PLANNING

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Course

DSA0101 Object Oriented Programming Using C++



ABSTRACT

- This project develops a **Smart Railway Reservation System** with an energy-efficient approach using **C++**.
- It enables users to **search trains, check availability, and book or cancel tickets** easily.
- The system optimizes train scheduling and resource usage to **minimize energy consumption**.
- Efficient data structures and file handling ensure **secure data management** and fast performance.
- The project aims to provide a **user-friendly, reliable, and sustainable** railway reservation solution.





INTRODUCTION



SMART RAILWAY RESERVATION WITH ENERGY-EFFICIENT



Smart and seamless reservation
anytime, anywhere.



Energy-efficient operations
for a sustainable future.



Optimized scheduling and real-time data
for better performance.



Better user experience with transparency,
speed and reliability.





Literature Review

Smart Railway Reservation with Efficient Energy Planning



 Year	 Research Paper	 Author(s)	 Key Findings	 Disadvantages	 Technologies Used
2020	Smart Railway Ticketing System Using IoT and Cloud Computing	R. Kumar, P. Trivedi, M. Naik	- IoT enables real-time seat availability and dynamic ticketing. - Reduces waiting time and improves passenger experience.	- Requires constant internet connectivity. - Higher initial infrastructure cost.	IoT, Cloud Computing, MySQL, RFID
2021	Energy-Efficient Train Scheduling Using Genetic Algorithm	S. Lee, J. Kim, H. Park	- Genetic algorithm optimizes train schedules to minimize energy consumption. - Provides better energy savings than traditional methods.	- Computational complexity increases with large datasets. - Requires accurate input parameters.	Genetic Algorithm, MATLAB, Simulation
2022	Integrated Framework for Smart Reservation and Energy Management in Railways	A. Mishra, D. Gupta, P. Sharma	- Combines reservation with energy management. - Improves resource utilization and operational efficiency.	- System integration is complex. - Data synchronization challenges.	MySQL, REST API, Energy Management System
2023	Predictive Energy Planning in Railways Using Machine Learning	T. Sathya, K. Rajesh, V. Priya	- ML models predict energy demand from historical train operations. - Helps in efficient energy allocation and cost reduction.	- Accuracy depends on quality of historical data. - Requires frequent model retraining.	Machine Learning, OpenCV, Data Processing
2024	Smart Railway Reservation System with Renewable Energy Integration	N. Verma, R. Sahu, P. Jain	- Integrates renewable energy sources with railway reservation. - Reduces carbon footprint and operational cost.	- Depends on renewable energy availability. - Installation and maintenance costs.	IoT Sensors, MySQL, Solar Energy System



OBJECTIVES

- Develop a **smart railway reservation system** for easy and efficient ticket booking.
- Provide **real-time seat availability** and automated reservation management.
- Manage **passenger and train information** efficiently in a centralized system.
- Monitor and analyze railway **energy consumption** based on train operations and passenger demand.
- **Optimize energy usage** by considering train occupancy, routes, and operational requirements.
- Use **data-driven planning** to reduce unnecessary energy consumption and improve efficiency.
- Integrate **reservation data with energy planning** to support smarter railway operations.
- Promote **sustainable railway transportation** by reducing energy wastage and operational costs.

EXISTING SYSTEM

- Reservation systems work **independently of energy management**.
- Ticket booking is **manual or semi-automated**, causing delays.
- Seat availability is **not updated in real-time**.
- **No integration** between passenger demand and train operations.
- Energy consumption is **not monitored or analyzed** effectively.
- Lack of **data-driven planning** leads to higher energy usage and costs.

PROPOSED SYSTEM

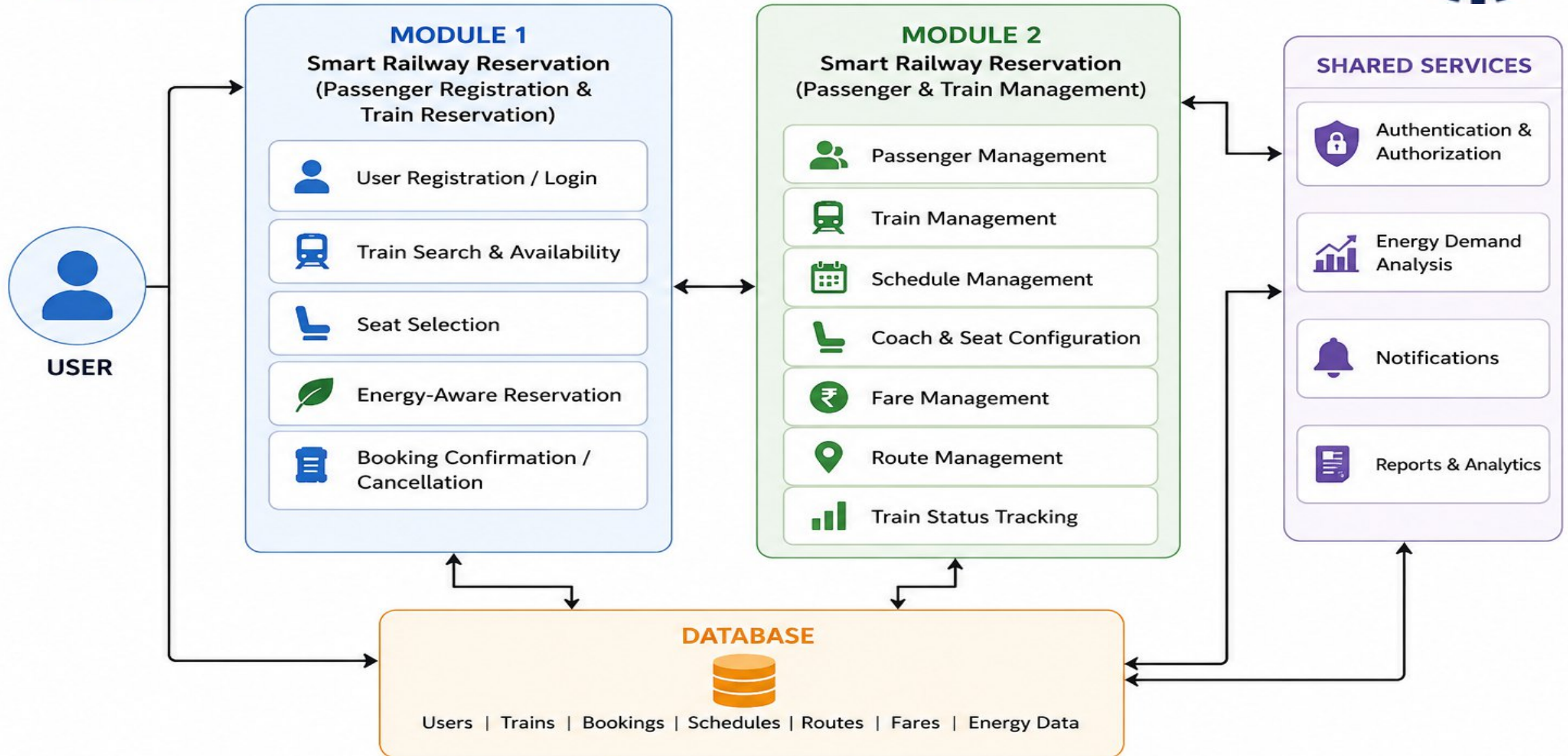
- An integrated **Smart Railway Reservation System** with **Efficient Energy Planning**.
- Provides **real-time** train schedules, seat availability, and instant booking.
- Automatically **updates seat availability** to avoid reservation conflicts.
- Integrates **passenger demand with train operations** for optimal resource utilization.
- Monitors and analyzes **energy consumption** based on occupancy, routes, and schedules.
- **Optimizes train scheduling and routing** to reduce energy usage and operational costs.
- Uses **data-driven planning** for efficient decision making and sustainable operations.
- Promotes a **user-friendly, reliable, and eco-friendly** railway system.



SYSTEM ARCHITECTURE



Smart Railway Reservation System





MODULE 1

SMART RAILWAY RESERVATION MODULE



Passenger Registration & Train Reservation

MODULE FEATURES

- User Registration / Login
- Train Search
- Train Availability
- Seat Selection
- Energy-Aware Reservation
- Booking Confirmation
- Cancellation

PROCESS FLOW

- Login / Register
- ↓
- Search Trains
- ↓
- Check Availability
- ↓
- Select Seat
- ↓
- Energy Demand Analysis
- ↓
- Confirm Booking



Supports efficient energy planning based on passenger demand.

INPUT

- Passenger Details
- Source & Destination
- Date
- Class
- Number of Passengers

PROCESS

- Search Trains
- Check Availability
- Select Seat
- Energy Analysis
- Confirm Booking

OUTPUT

- Confirmed Reservation
- Estimated Energy Demand Information

TECHNOLOGIES USED

C++



JDBC

```

1 #include <iostream>
2 using namespace std;
3 int main() {
4     string name, train;
5     int seats;
6     cout << "Enter Passenger Name: ";
7     cin >> name;
8     cout << "Enter Train Name: ";
9     cin >> train;
10    cout << "Enter Seats: ";
11    cin >> seats;
12    if (seats > 0 && seats <= 100) {
13        cout << "\nBooking Confirmed";
14        cout << "\nPassenger: " << name;
15        cout << "\nTrain: " << train;
16        cout << "\nSeats: " << seats;
17        cout << "\nFare: Rs." << seats * 450;
18    } else {
19        cout << "Invalid seats";
20    }
21    return 0;
22 }

```

Output

```

Enter Passenger Name: ABI
Enter Train Name: 56
Enter Seats: 7

```

```

Booking Confirmed
Passenger: ABI
Train: 56
Seats: 7
Fare: Rs.3150

```

```

=== Code Execution Successful ===

```




MODULE 2

SMART RAILWAY RESERVATION MODULE



Passenger & Train Management

MODULE FEATURES

-  Passenger Management
-  Train Management
-  Schedule Management
-  Coach & Seat Configuration
-  Fare Management
-  Route Management
-  Train Status Tracking

PROCESS FLOW

- Add / Update Passenger
- ↓
- Add / Update Train
- ↓
- Set Train Schedule
- ↓
- Configure Coaches & Seats
- ↓
- Set Fare Details
- ↓
- Define Routes
- ↓
- Track Train Status



Ensures accurate management of trains and passengers for smooth operations and efficient planning.

INPUT

-  Passenger Details
-  Train Details
-  Schedule Information
-  Coach & Seat Layout
-  Fare Details
-  Route Information

PROCESS

-  Add / Update Passenger
-  Add / Update Train
-  Schedule Trains
-  Configure Coaches & Seats
-  Set Fare
-  Manage Routes
-  Track Status

OUTPUT

-  Passenger Database
-  Train Database
-  Schedule & Route Information
-  Seat Availability Information
-  Fare Information
-  Train Status Reports

TECHNOLOGIES USED

C++



JDBC

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int passengers;
5     double energy;
6     cout << "Enter Number of Passengers: ";
7     cin >> passengers;
8     energy = passengers * 2.5;
9     cout << "\nEnergy Usage: " << energy << " units";
10    if (passengers >= 50)
11        cout << "\nStatus: Efficient";
12    else
13        cout << "\nStatus: Normal";
14    return 0;
15 }
```

Output

Enter Number of Passengers: 30

Energy Usage: 75 units

Status: Normal

=== Code Execution Successful ===

Output

Enter Number of Passengers: 60

Energy Usage: 150 units

Status: Efficient

=== Code Execution Successful ===



CONCLUSION



The system provides a **seamless and efficient** platform for passenger and train management.



Energy-aware reservation supports **efficient energy planning** based on passenger demand.



Ensures **accurate** data management, real-time availability, and **secure** bookings.



Modular design enables **easy maintenance, scalability,** and future enhancements.



Built using **C++**, ensuring high performance, reliability, and efficient resource utilization.





REFERENCES

Smart Railway Reservation with Efficient Energy Planning



01

R. Kumar, P. Trivedi and M. Naik, **"Smart Railway Ticketing System Using IoT and Cloud Computing,"** 2020.



02

S. Lee, J. Kim and H. Park, **"Energy-Efficient Train Scheduling Using Genetic Algorithm,"** 2021.



03

A. Mishra, D. Gupta and P. Sharma, **"Integrated Framework for Smart Reservation and Energy Management in Railways,"** 2022.



04

T. Sathya, K. Rajesh and V. Priya, **"Predictive Energy Planning in Railways Using Machine Learning,"** 2023.



05

N. Verma, R. Sahu and P. Jain, **"Smart Railway Reservation System with Renewable Energy Integration,"** 2024.





THANK YOU

Smart Railway Reservation
with Efficient Energy Planning



Thank you for your time and attention.
Any Questions?

